

An Analysis of Ludo Board Game Play on Smartphones

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ABSTRACT

From sports to party games, almost every kind of game has been adapted into a digital video game format. While previous research has studied player motivations and experiences for certain categories of digital games, there has yet to be such a study on digital board games, especially in the modern context of smartphone apps. To address this, we conduct a case study of a popular board game, Ludo, to understand players' opinions of its digital adaptation. For this, we study the functionality and user reviews of nine popular Ludo apps, to assess player opinions of how traditional gameplay has been re-imagined. Based upon our analysis, we conclude with recommendations for improving Ludo apps and other apps, based on random chance board games.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

Mobile Apps, Board Games, User Reviews

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1 INTRODUCTION

Games are an important part of human life and are common to all cultures across the globe. Over the years, almost every type of game has been adapted into a digital format, starting with Pong, one of the first video games which was based on table tennis. Previous work has investigated player opinions of role-playing [3] and card games [28]. However, there is little research investigating user opinions of traditional board games that have been transposed

into digital formats. Existing studies have mostly focused on analysis of quantitative data [2, 11, 14, 19] with little attention to a qualitative analysis of players opinions. Moreover, these studies are concerned with desktop games, rather than smartphone apps which now account for more than 50% of the current video games market [31].

Since board games are a very broad category with many subcategories within them, it is not possible to do full justice to this entire domain in just one research article. Hence, we present a case study for a specific board game to gain insights into the usability of its mobile adaptations, some of which may possibly generalise to other board games. For this work, we focus on the game Ludo, a game derived from Pachisi, a traditional board game from India. Ludo is a two to four-player game consisting of elements common to many other board games such as token racing, dice-based movement and a cross-shaped board (Figure 1). It became popular particularly in the US throughout the 60s and up until recently with the Parker Brothers board game 'Parcheesi'. The game is widespread in the Indian subcontinent and has been said to act as a 'social lubricant' having the ability to promote interactions between diverse social groups [7]. Due to the recent COVID-19 pandemic and subsequent lockdown, Ludo has seen a significant spike in popularity with 12 Ludo mobile apps being among the top 200 free games on the Indian Play Store in March 2020 [26]. Additionally, the most popular app, Ludo King was the 5th most downloaded mobile game worldwide in April 2020 [4].

2 ABOUT THE GAME

In this section, we trace the origin of the game and describe how to play it. While different variations and house rules exist, we present the rules followed by the app-based Ludo games that we reviewed for this study which are consistent across all the apps.

2.1 History

The origin of the modern Ludo played on mobile apps can be traced back to the game Pachisi. While the exact origins of Pachisi are unclear, games very similar to it have been played in different parts of the world as early as 220-65 AD [15]. Evidence of Pachisi being played in the 6th century exists in temples of the Ellora Caves in Maharashtra, India, where the game board was found to be carved in stone [13]. The Hindu epic Mahabharata [30] mentions a game Chaupar which is very similar to Pachisi. In the 16th Century, Mughal Emperor Akbar used to play the game with his court being

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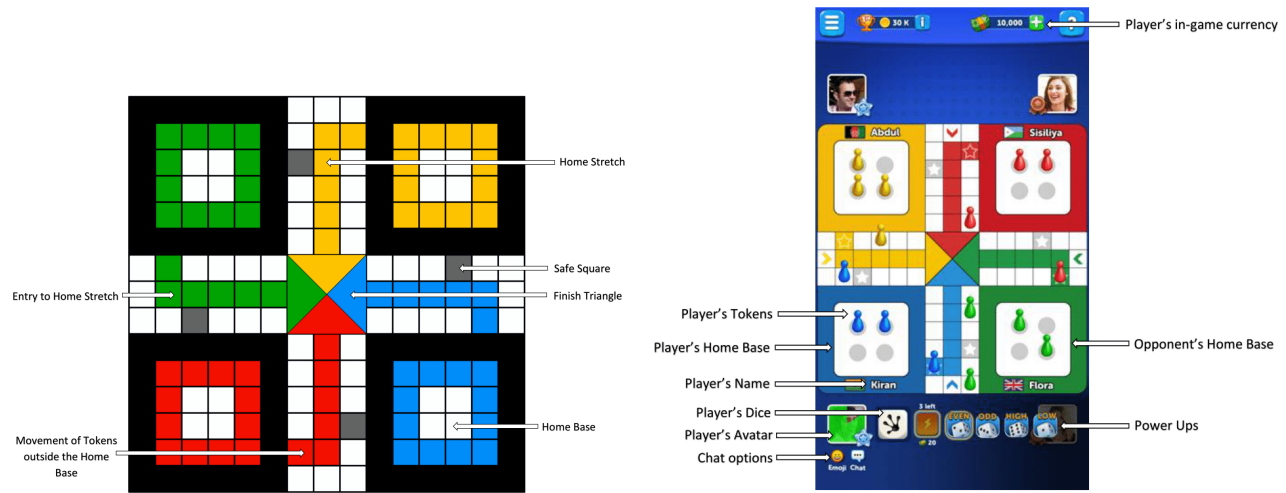


Figure 1: An example of a Ludo board (left), Screenshot of a game in progress in the Ludo Club app (right)

the game board and his slaves acting as the game pieces while wearing clothing which represented the four-game colours [25]. The modern Ludo which is most widely played is attributed to Alfred Collier, who patented the game in the late 19th-century [6]. Other games similar to Ludo were developed in the 20th-century such as Mensch ärgere Dich nicht [22], Sorry! [16], Aeroplane Chess [9], Uckers [20] and many more.

The Ludo board consists of four large squares at the corners coloured red, green, yellow and blue (Figure 1). These squares may be referred to as the 'home base' or 'yard' [8]. The game track is cross-shaped consisting of three columns. The central column is coloured corresponding to the 'home base' next to it and known as the 'home stretch' [22]. The centre of the cross is a square made up of four triangles corresponding to the four colours. This is called the 'finish triangle'. In addition, the track is scattered with four squares marked with a different colour or symbol, known as safe squares.

2.2 The Rules

Ludo can be played between two to four players individually, with each player taking one of the four colours on the board. All players have four tokens with the colour being the same as their home base. These are placed in the home base at the beginning of the game. The order of the turns is decided randomly or based on the throw of the dice. The objective of the game is to be the first player to get all tokens inside the finish triangle. To reach the finish triangle, a token has to complete one round of the track, after which it can enter the home stretch corresponding to its colour. The game may stop when one player wins or may continue to determine other players' rankings.

To move a token from the home base to the game track, a player must roll a six. Rolling a six also awards a player with an additional throw except in the occurrence of three consecutive sixes which leads to the turn getting cancelled. Tokens move clockwise around the track with pieces advancing each square equal to the number thrown on the dice. If a player has some tokens on the track and some in the home base, they can choose whether to progress

through the squares or bring a token from the home base to the track. If no legal move is possible based on the number on the dice, the turn is forfeited.

If a player's token is moved to a square which is already occupied by another player's token, the occupying token is sent back to the home base. This event is known as 'capturing' or 'taking'. If two or more tokens of the same colour are on a square, they can only be captured if an equal number of tokens of another colour move to that square. Pieces are also safe when they are on coloured or specially marked 'safe' squares.

3 RESEARCH METHODS

To understand Ludo player opinions and experiences, we conducted a qualitative analysis of Ludo app reviews. This approach is similar to previous works [1, 2, 10, 23] which used smartphone app reviews to gain insights on their usage. We selected all Ludo apps in the top 100 free games category on both the Android Play Store and iOS App Store in India during March 2020, which was the peak of the game's popularity. These included: *Ludo King*, *Ludo by Yarsa Games*, *Ludo SuperStar*, *Ludo Club*, *Ludo All Star*, *Ludo Talent*, *Ludo Master*, *Yalla Ludo*, *Ludo Star* and *Ludo Star 2*. Due to Ludo Star's features being a subset of Ludo Star 2's, we decided to focus only on Ludo Star 2 instead of both, leaving us with 9 apps. All 9 of these apps are available on the Play Store and 6 available on the App Store. The study design received IRB approval.

3.1 Functionality Review

To better understand user reviews and opinions, we first needed to gain an understanding of the features available in each app. This also helped us identify certain unique features which users did not discuss in the reviews. We catalogued the features and functionality of each app including different game modes, settings options, in-game stores and other features. This analysis was conducted independently by two of the authors who then combined their findings by exchanging written notes.

3.2 User Review Analysis

Next, we conducted a thematic analysis of 450 (9 X 50) Ludo app reviews. First, personally identifiable information was removed from the reviews and then, they were analysed using open coding [27]. Two of the authors independently coded all reviews and then reviewed the codes over a series of discussions. The open-source tool Taguette¹ was used for coding the reviews.

For each app, we selected the top 10 reviews from each star rating (from 1-5) to sample a variety of opinions for a total of 50 reviews per app. If the app was available on both iOS and Android then half of the reviews were downloaded from the App Store and the other half from the Play Store. We used the default sorting of both the app marketplaces which was sort by relevance and sort by recent for the Play Store and App Store respectively. Both Gebauer et al. [12] and Vasa et al. [29] have determined that mobile app reviews are usually short. Sorting by relevance helped us compensate for this effect in case of the Play Store by providing recent reviews that are also longer and detailed². Since there was no such option for the App Store, we ignored any reviews which did not have at least one complete sentence for the iOS apps.

4 FINDINGS

4.1 Functionality Review

All of the 9 Ludo Games have a common game view consisting of the entire game board positioned such that the player's colour square is at the bottom left corner. The player's avatar, along with the dice is also present at the bottom left below the game board. The dice is rolled by tapping on its icon when it appears on the screen during a player's turn. A rolling animation is triggered with one of the six faces being displayed. Ludo by Yarsa Games had an alternative rolling interaction available which aimed to emulate the action of shaking the dice in a player's hand before a roll, by triggering the roll-on shaking the mobile phone. This feature, though distinct, was not mentioned in any of the app's reviews. 4/9 games provided skins for the dice bought using in-game currency while some had a single dice design which changed its colour corresponding to the player whose turn it is. During online games, an auto roll mode is available in all nine apps, which takes control over a player when their connection is broken, such that the game is not interrupted.

Tokens are controlled by tapping the token to be moved based on the dice number. All 9 games had a Quality of life feature which automatically moved a token if only one legal move was possible. The ability to choose the colour of the token was present in 9/9 games during offline play and 3/9 during online play. Similar to dice skins, 3/9 games provided the option for token skins. Skins for the board and overall game were also present in some apps. 7/9 apps had alternate modes consisting of variations on the classic ludo rules. While each app had its own unique variation, the common theme across all of them was that they were aimed at reducing the length of the game. Shorter game times are designed into the alternate modes by:

- requiring only two tokens to be moved to the finish triangle (Ludo All Star, Ludo Master)

- adding power-ups to certain squares to increase movement speed (Ludo Talent)
- each player has their own dice which is rolled independently in a race to end the game (Ludo King)
- changing the game's objective to cutting the opponent's tokens instead of reaching the finish triangle (Ludo Club, Ludo All Star)

4.2 User Review Analysis

From our analysis of user reviews, three categories of user opinions emerged: Technical Features, Social Aspects and Player Motivations. Since app reviews are known to often contain misspelt words, slangs and short forms [24], the user quotes in this section have been through a grammar and spelling check for readability.

4.2.1 Technical Features.

Users have strong opinions about features in the apps and were especially vocal about their dissatisfaction with how the rolling of the dice is implemented. Ludo is supposed to be a "game of luck" (Review #131, 4★) as one reviewer put it. However, a majority of the reviews across all nine apps mention that the **dice is "biased"** and that the probability for each face of the dice is not always equal. Reviewers mentioned that the following occurrences were common: when a specific number is needed to cut another player's token the exact number needed lands on the dice; when a player is a single number away from winning the number is not rolled on the dice while the opponents get large numbers to make them move faster; increased number of sixes when all of a player's tokens are in the home base; consecutive high numbers for some players with consecutive low numbers for others.

While there is no way to verify that dice probabilities are being manipulated without access to the apps' codebases, the authors also observed the above-mentioned phenomenon very often in informal testing. Reviewers speculate that the **dice algorithm** is set to make the games 'close' with the aim of making the experience exciting: "for some reason (to make games more close) the creators decided to include their 'hand' in our matches" (Review #173, 3★). Other reviewers guess that the developers wanted to encourage players to spend money on in-app purchases: "It also lets you buy (in-app currency) at cheaper costs so that you feel happy but makes you lose it even faster and you are caught in the cycle of futile expenditure" (Review #196, 5★). Instead of making the game better, however, reviews state that dealing with the biased dice made them "so frustrated that I uninstalled it" (Review #233, 4★), "so angry that I was about to break my phone" (Review #168, 2★) and this feature has "made quite a lot of people unhappy and has pulverised the joy out of it" (Review #106, 1★).

8/9 apps had a **levelling system** where players earned experience points and received rewards for reaching higher levels. The was criticised in many reviews saying that game developers increasingly rigged the dice against users when they reached higher levels of the game. While increasing levels do correspond to more difficulty, users felt that changing the random chance property of the game was not the way: "As you start playing regularly, you are likely to feel frustrated and feel that something is not right because you will start to struggle even to get the token out. Adding to that

¹<https://www.taguette.org/>

²<https://intercom.help/localclarity/en/articles/3044081-what-are-the-factors-for-a-google-review-to-be-marked-most-relevant>

your opponent will get whatever crazy combinations they need to kill your token" (Review #159, 1★).

Other than general probability manipulation in games, some apps also have direct ways to manipulate the dice probability in the form of **power-ups**. The most common is the ability to re-roll the dice by spending a small amount of in-game currency. Others include rolling dice with only even numbers or with specific numbers. Reviewers commented that these power-ups *"seem to take away the fairness in the game"* (Review #110, 1★) and using them amounts to cheating. *"Undo feature is cheating"* (Review #36, 4★).

While there were very few reviews directly appreciating **cosmetic customization**, there were many requests for these features in apps where customization options were absent. While the *"concept of types of dice and themes are interesting"* (Review #436, 4★), users express a desire to customize their game personally where they are able to choose specific colours. They said that developers should *"add the option to choose the pawn (token) colour"* (Review #25, 3★) and to *"give us some new themes for boards"* (Review #221, 3★). Other reviews mentioned that the Ludo board *"need(s) more improvement"* (Review #68, 4★), but did not specify their exact issue with the appearance of the board. *"Please give theme settings so that it will look more beautiful"* (Review #387, 4★). Another suggestion related to customization was to *"introduce an Indian version of chokad, pachisi, or an old Indian theme, thus keeping the very essence of an Indian origin"* (Review #182, 4★).

Accessibility features were requested in some reviews, for example: *"I am blind and I wish to play ludo in your application"* (Review #142, 3★). Another reviewer wanted to be able to switch to their dominant hand: *"An option to put our base on the right side to make right-hand playing more comfortable"* (Review #250, 5★). This could indicate that the apps lacked features to make the game inclusive despite the possibility of leveraging the digital platform to support such functionality.

The attempt to create **alternate modes** to reduce game time is in line with reviewer comments such as *"This game is very boring"* (Review #54, 1★) and *"it gets dragged over 25 mins for 4 people!"* (Review #182, 4★). These alternate modes were very well received by players, for example: *"I like the Raid mode a lot"* (Review #36, 4★). *"No doubt the magic option is amazing"* (Review #363, 2★). They appreciated that there are *"more options to play other than Classic"* (Review #143, 5★), felt that this *"enhances the classic board game"* (Review #346, 5★) and acknowledged the lesser game length *"If you are in hurry, you can try Raid mode"* (Review #45, 5★).

Despite primarily being Ludo games, certain apps such as Ludo King and Ludo Star 2 try to differentiate themselves by providing **additional games to play** alternatives. Players have appreciated being able to play other childhood games, and have requested these games in apps lacking this feature. *"I liked that the name is Ludo Master, but in the game, you can also play snake and ladders"* (Review #242, 5★). *"If possible, add snake ladder in the same game"* (Review #92, 5★). Allowing players to access different rules and other games provides variety in gameplay and is likely aimed at retaining players as active users.

4.2.2 Social Aspects.

Apps that support online multiplayer games include the option to communicate with players through different mediums such as

text, emoji, voice chat and stickers. Many players noted that the ability to play and interact with others across the platform allowed them to **engage with friends** *"The best part for me is that you can play it with your friends."* (Review #78, 3★). Similar to the board game, the digital game is also a *"social lubricant"*[7] that brings people together. For instance, *"Every day you will make new friends"* (Review #393, 5★) *"Love this game, very entertaining, especially at this time you can meet new friends and play with them"* (Review #141, 5★).

People across different countries have also been brought together by playing Ludo online: *"(It) helps in making friends in different countries"* (Review #345, 5★). In addition, Ludo games have served as a **means for families to bond**, allowing them to spend time with their loved ones: *"I loved this game because this is amazing in my friends and family circle"* (Review #172, 3★). A few reviews mention specific family members like fathers, daughters, and sisters as game partners. *"My daughter and I enjoy playing this game"* (Review #193, 5★) and *"I am playing with my sisters and with my dad, and it's so much fun"* (Review #98, 5★).

Reviewers also requested feature changes with most in relation to **support for social interactions**. For instance, a highly requested feature was to increase the maximum number of players from 4 to 6: *"6 players multiplayer ludo will be the best, you know, for a big group of friends"* (Review #347, 5★). A variation of this request was to allow spectators in a game: *"add a feature for another person who is not playing to watch"* (Review #172, 3★). Some reviewers wanted to further engage with other players they were randomly matched to through a rematch option: *"Why is there no rematch option after the game ends? So we can take revenge"* (Review #426, 3★). Also, in-app chat outside the game was requested: *"please add chat with an opponent feature in the lobby"* (Review #272, 3★). In addition, an option to choose opponents was suggested: *"There should be an option for choosing our opponent"* (Review #175, 3★).

On the darker side, **misuse of social features** was mentioned across reviews: *"some guys are using the gift-emojis to harass female players"* (Review #188, 4★), and: *"I am subjected to extremely abusive language"* (Review #142, 5★). Additionally, the ability to block, report and mute other players were demanded in many reviews: *"Please make a provision to block and report the players who abuse or use foul language while playing"* (Review #150, 5★) *"You can provide an option to disable or enable the voice of another player"* (Review #189, 4★). Here, it seems that most apps did not have checks in place for harassment and cyberbullying.

4.2.3 Player Motivation.

Ludo is a household game in India. It is a common sight for friends and family to bond over the board game. *"I have loved playing ludo since my childhood days"* (Review #315, 2★) *"I have been playing since I was 8 years old"* (Review #318, 2★). When the game was introduced online, it reminded people of their childhood days and allowed them to bond with others who had a similar interest in the game. *"It is the best game to share interests and childhood memories with your friends"* (Review #243, 5★) *"This is the platform that provides you with the space to connect with your family and friends and play ludo with them and recall your childhood"* (Review #249, 5★)

Another motivation among many users to play the game has been to **pass time**. The game has served as a *"very nice time pass*

with friends and family” (Review #396, 5★). Reviews also mentioned that the game helped them to relax: *“It’s a good time pass and stress buster”* (Review #313, 2★). A frequently mentioned aspect of this is being able to connect with other people while passing time: *“My wife and I are in love with this game. All the family members play it together in our free time”* (Review #341, 5★). *“It is a good game to spend time with friends”* (Review #114, 2★). The ability to connect with friends and family through online multiplayer games was considered important in the reviews.

The effects of the popularity of Ludo apps during the **Coronavirus Lockdown** was also evident in the reviews. During the lockdown, Ludo apps seemed to help players pass the time: *“I had downloaded the game during lockdown when watching TV or movies was boring, so in between, this was a welcome break to pass your time”* (Review #41, 5★). The reviews reveal that many players used the game as a stress buster: *“During these lockdown days, Ludo is the one app that relieves all stress and brings joy and fun.”* (Review #187, 4★). Some apps introduced Coronavirus related content: *“They give weekly tasks with some tempting dice saying ‘saluting corona warriors during the lockdown’”* (Review #108, 1★). Ludo mobile phone games have also provided a platform for users to be social while practising social distancing. *“Love this game, very entertaining, especially during this time; you can meet new friends and play with them”* (Review #141, 5★). *“This is a great game to play with family, especially in times of the Coronavirus”* (Review #146, 5★).

Some reviews compared the mobile apps to the board game: *“This game will make you forget the board”* (Review #198, 5★), and *“I’ve always liked playing the Ludo board game but never thought it would be so much fun playing it on a cellphone”* (Review #136, 4★). Ludo once required a large box with a big board and tokens. Now it has become much more convenient and portable: *“It has mobility. We can carry it from one place to another”* (Review #192, 5★), and: *“Back then it used to be on hardbound paper-boards. Now, it’s on our phones”* (Review #315, 2★). While physical board games can be cumbersome to carry, mobile apps provide an option to **play on the go**.

5 DISCUSSION

In the section below, we describe how our findings can be extended further and how our analysis relates to prior research on app reviews.

5.1 Implications for Board Game Apps

While our study is only focused on apps for Ludo, we believe that some of the players’ criticisms and suggestions are also applicable for designers to leverage for other board game apps. Therefore, we present below some general recommendations that may improve board game apps, especially those involving random chance.

- Randomizing devices such as dice or spinning tops should not be manipulated even if the aim is to improve the game experience. Players are able to easily catch on to these causing them to lose trust in the game.
- Players love to customize their game elements and it is likely that unique skins are a source of personal expression. Increasing options for cosmetic customization is a better avenue to encourage profit generating in-app purchases compared to features that modify probabilities (such as power ups).

- Given that a large number of house rules and variations exist for older board games, players are not purists when it comes to how games are to be played. Variety in gameplay is appreciated by users and is important for increasing the lifespan of an app. This strategy to increase longevity has also been used by some board game manufacturers when providing reversible double-sided boards supporting multiple games. Another reason to provide variations is to tweak the rules based on user needs such as short game lengths.
- Actual board games are rarely played with just a single player and interacting with others is a major part of the experience. Board game apps should try to emulate this aspect by providing ample opportunities for socializing with other players. However, apps should ensure that communication between players is never an unpleasant experience by having checks in place for profanity, bullying and harassment. While this recommendation is not specific to board game apps, we mention it as it was a common pain point among players.

5.2 Observed App Review Phenomenon

User reviews are a direct indicator of how a product or application is perceived by its users and thus is a rich source of information for improving them. Pagano and Maalej [21] state that users often complain about removed or changed features and this fact is also evident in our review set. In our case, users observe missing features when they switched apps and complained about their absence. Lin et al. showed that users value game design over software quality (i.e., the number of bugs in a game) [18]. This phenomenon is noticeable in the case of Ludo games with users writing reviews in appreciation of the UI, praise of graphics, and overall gameplay. The “Brag and Moan” model by Hu et al. assumes users only write reviews when they are very satisfied or very disgruntled [17]. Ludo app reviews did not usually follow this model as even the 4 or 5 reviews mentioned being dissatisfied with features related to dice probability. Chen and Lui observe that top-ranked apps are not necessarily the highest rated [5]. Among Ludo apps, Ludo King (Play Store: 4.1/5, App Store: 3.8/5) is the most popular app but the highest rated app is actually Ludo Club (Play Store: 4.4/5, App Store: 4.3/5).

6 CONCLUSION AND FUTURE WORK

We conducted this study on smartphone apps for a specific board game, Ludo. We find that users especially appreciate cosmetic customization as well as different ways to interact with their online opponents. Changing the random chance nature of the game in an attempt to make gameplay more exciting received backlash, however, adding variations to the rules led to positive feedback. We plan to further extend this work by studying other distinct categories of board games such as strategy or word games with the aim of creating a general framework for designing board game apps.

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